Physical Chemistry of Materials

Lecture 08

Grand Canonical Ensemble

Last Class

Intramolecular Interactions

Interaction type	Distance dependence of potential energy	Typical energy/ ($kJ\,mol^{-1}$)	Comment
Ion–ion	$1/r$	250	Only between ions
Hydrogen bond		20	Occurs in X–H···Y, where X, Y = N, O, or F
Ion–dipole	$1/r^2$	15	
Dipole–dipole	$1/r^3$	2	Between stationary polar molecules
	$1/r^6$	0.3	Between rotating polar molecules
London (dispersion)	$1/r^6$	2	Between all types of molecules and ions

(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th Ed., Table 14B.1)

Last Class

Phase Transitions

- Phase transitions: ordered ↔ disordered
- *e.g.* liquid-gas transition, magnetic transition

- Competition between energy and entropy
- Interactions are required!

Last Class

(Monatomic) Gas Models

- Ideal gas
  - Energy $E = \sum_i (p_i^2/2m)$
  - Partition function

$$Z = \frac{1}{N!\,h^{3N}} \iint e^{-\sum_i p_i^2/2mk_\mathrm{B}T}\,\mathrm{d}^N\vec{r}\,\mathrm{d}^N\vec{p} = \frac{V^N}{N!}\left(\frac{2\pi m k_\mathrm{B} T}{h^2}\right)^{3N/2}$$

- Real gas
  - Energy $E = \sum_i (p_i^2/2m) + \sum_{i<j} E_\mathrm{pot}(r_{ij})$

Last Class

Real-Gas Partition Function

$$E = \sum_i (p_i^2/2m) + \sum_{i<j} E_{\text{pot}}(r_{ij})$$

$$Z = \frac{1}{N!}\left(\frac{2\pi m k_{\text{B}} T}{h^2}\right)^{3N/2} \int e^{-\sum_{i<j} E_{\text{pot}}(r_{ij})/k_{\text{B}}T}\, \mathrm{d}^N \vec{r}$$

$$Z = \frac{V^N}{N!}\left(\frac{2\pi m k_{\text{B}} T}{h^2}\right)^{3N/2} \left(1 + \frac{N^2}{2V} I\right)$$

where

$$I = \int \left(e^{-E_{\text{pot}}(r)/k_{\text{B}}T} - 1\right) \mathrm{d}\vec{r}$$

Evaluation of I

- Sutherland potential: hard sphere + dispersion force

$$E_{\mathrm{pot}}(r) = \begin{cases} +\infty, & r < r_0 \\ -\epsilon_0 \left(\frac{r_0}{r}\right)^6, & r \geq r_0 \end{cases}$$

$$I = -\frac{4}{3}\pi r_0^3 \left(1 - \frac{\epsilon_0}{k_{\mathrm{B}} T}\right)$$

By calculating $p = -(\partial A/\partial V)_{T,n}$,

one can obtain the van der Waals equation of state

(Thermodynamic) Ensemble

Idealized group of a large number of virtual copies

(sometimes infinitely many) of a system

Microcanonical ensemble

Canonical ensemble

Grand canonical ensemble

⋮

Grand Canonical Ensemble

Thermodynamic ensemble of an open system

- Conserved quantities: μ, V, T
- Microstates may have different energies
- All microstates are NOT equally probable
- Constrained optimization of entropy!

Review: Canonical Ensemble

$$S/k_{\mathrm{B}} = -\sum_i p_i \ln p_i$$

$$\sum_i p_i - 1 = 0$$

We can introduce one more constraint: $\sum_i n_i E_i = \mathcal{N} U$

As the constraint implies $\sum_i p_i E_i - U = 0$,

$$\frac{\partial[S/k_{\mathrm{B}} - \lambda(\sum_i p_i - 1) - \beta(\sum_i p_i E_i - U)]}{\partial p_i} = 0$$

$$p_i = e^{-1-\lambda-\beta E_i}$$

Particle Number Conservation

$$S/k_{\mathrm{B}} = -\sum_i p_i \ln p_i$$

- Constraint 1: $\sum_i p_i - 1 = 0$
- Constraint 2: $\sum_i p_i E_i - U = 0$
- Additional constraint: $\sum_i n_i N_i = N_{\mathrm{total}}$, or

$$\sum_i p_i N_i - \frac{N_{\mathrm{total}}}{\mathcal{N}} = 0$$

Applying Lagrange's multiplier method,

$$p_i = e^{-1-\lambda-\beta E_i-\alpha N_i}$$

Grand Partition Function

$$p_i = e^{-1-\lambda-\beta E_i-\alpha N_i}$$

From the normalization constraint,

$$\sum_i p_i = e^{-1-\lambda} \sum_i e^{-\beta E_i-\alpha N_i} = 1$$

Define the grand partition function $\mathcal{Z}$:

$$\mathcal{Z} = e^{1+\lambda} = \sum_i \underbrace{e^{-\beta E_i-\alpha N_i}}_{\text{Gibbs factor}}$$

Back to Entropy

$$p_i = e^{-\beta E_i - \alpha N_i}/\mathcal{Z}$$

$$S/k_\mathrm{B} = -\sum_i p_i \ln p_i = -\sum_i p_i \ln\left(\frac{e^{-\beta E_i - \alpha N_i}}{\mathcal{Z}}\right)$$

$$S/k_\mathrm{B} = -\sum_i p_i(-\beta E_i - \alpha N_i - \ln \mathcal{Z})$$

$$S/k_\mathrm{B} = \beta \sum_i p_i E_i + \alpha \sum_i p_i N_i + \ln \mathcal{Z}$$

$$\therefore \; S/k_\mathrm{B} = \beta U + \alpha \langle N \rangle + \ln \mathcal{Z}$$

Bridge to Thermodynamics

$$S = k_{\mathrm{B}}\beta U + k_{\mathrm{B}}\alpha\langle N\rangle + k_{\mathrm{B}}\ln\mathcal{Z}$$

$$S = k_{\mathrm{B}}\beta U + N_{\mathrm{A}}k_{\mathrm{B}}\alpha n + k_{\mathrm{B}}\ln\mathcal{Z}$$

From thermodynamics, we know

$$\left(\frac{\partial S}{\partial U}\right)_{V,n} = \frac{1}{T}, \qquad \left(\frac{\partial S}{\partial n}\right)_{U,V} = -\frac{\mu}{T}$$

Hence,

$$\beta = \frac{1}{k_{\mathrm{B}}T}, \qquad \alpha = -\frac{\mu}{N_{\mathrm{A}}k_{\mathrm{B}}T} = -\frac{\tilde{\mu}}{k_{\mathrm{B}}T}$$

Grand Potential

$$S = \frac{U}{T} - \frac{\mu n}{T} + k_{\mathrm{B}} \ln \mathcal{Z}$$

$$-k_{\mathrm{B}} T \ln \mathcal{Z} = U - TS - \mu n = A - \mu n$$

(Legendre transform of A)

Define the grand potential $\phi = -k_{\mathrm{B}} T \ln \mathcal{Z}$;

$$\mathrm{d}\phi = \mathrm{d}A - \mathrm{d}(\mu n) = -S\mathrm{d}T - p\mathrm{d}V - n\mathrm{d}\mu$$

$$\therefore \phi = \phi(T, V, \mu)$$

Derived Functions

$$\phi = -k_{\mathrm{B}} T \ln \mathcal{Z}$$

Using $\mathrm{d}\phi = -S\mathrm{d}T - p\mathrm{d}V - n\mathrm{d}\mu$,

$$S = -\left(\frac{\partial \phi}{\partial T}\right)_{V,\mu}, \qquad p = -\left(\frac{\partial \phi}{\partial V}\right)_{T,\mu}, \qquad n = -\left(\frac{\partial \phi}{\partial \mu}\right)_{T,V}$$

Summary

For a grand canonical ensemble, the probability to find the system in a microstate i (with energy E_i and particle number N_i) is

$$p_i = \frac{e^{-\beta E_i - \alpha N_i}}{\mathcal{Z}} = \frac{e^{-E_i/k_\mathrm{B}T + \tilde{\mu} N_i/k_\mathrm{B}T}}{\mathcal{Z}}$$

where the grand partition function $\mathcal{Z} = \sum_i e^{-\beta E_i - \alpha N_i}$

Also, the grand potential is given by

$$\phi = -k_\mathrm{B} T \ln \mathcal{Z}$$

Average Energy

$$U = \langle E \rangle = \sum_i p_i E_i = \sum_i \frac{E_i e^{-\beta E_i - \alpha N_i}}{\mathcal{Z}}$$

However,

$$\frac{\partial \ln \mathcal{Z}}{\partial \beta} = \frac{1}{\mathcal{Z}} \frac{\partial \mathcal{Z}}{\partial \beta} = -\frac{1}{\mathcal{Z}} \sum_i E_i e^{-\beta E_i - \alpha N_i}$$

Hence,

$$U = -\frac{\partial \ln \mathcal{Z}}{\partial \beta}$$

Average Particle Number

$$\langle N \rangle = \sum_i p_i N_i = \sum_i \frac{N_i e^{-\beta E_i - \alpha N_i}}{\mathcal{Z}}$$

However,

$$\frac{\partial \ln \mathcal{Z}}{\partial \alpha} = \frac{1}{\mathcal{Z}} \frac{\partial \mathcal{Z}}{\partial \alpha} = -\frac{1}{\mathcal{Z}} \sum_i N_i e^{-\beta E_i - \alpha N_i}$$

Hence,

$$\langle N \rangle = -\frac{\partial \ln \mathcal{Z}}{\partial \alpha}$$

Rearrangement

$$\mathcal{Z} = \sum_i e^{-\beta E_i - \alpha N_i}$$

By grouping by the particle number, we can rearrange the summation as follows:

$$\mathcal{Z} = \sum_{N=0}^{\infty} e^{-\alpha N} \left(\sum_s e^{-\beta E_s(N)} \right) = \sum_N \left(e^{\mu/RT} \right)^N Z(N)$$

What is $e^{\mu/RT}$?

For a “real” gas mixture, we have

$$\mu = \mu^{\ominus} + RT \ln \frac{f}{p^{\ominus}}$$

where f is the fugacity.

$$e^{\mu/RT} = \frac{e^{\mu^{\ominus}/RT}}{p^{\ominus}} f \propto f$$

In statistical mechanics, we simply call $e^{\mu/RT}$ the fugacity.

Calculation of $\mathcal{Z}$

$$\mathcal{Z} = \sum_N \left(e^{\mu/RT}\right)^N Z(N)$$

- Distinguishable particles: $Z(N) = z^N$

$$\mathcal{Z} = \sum_N \left(e^{\mu/RT}\right)^N z^N = \sum_N \left(ze^{\mu/RT}\right)^N$$

- Indistinguishable particles: $Z(N) = \frac{1}{N!} z^N$

$$\mathcal{Z} = \sum_N \left(e^{\mu/RT}\right)^N \frac{z^N}{N!} = \sum_N \frac{1}{N!} \left(ze^{\mu/RT}\right)^N = \exp\left(ze^{\mu/RT}\right)$$

Example: 3D Ideal Gas

$$z = V\left(\frac{2\pi m k_{\mathrm{B}} T}{h^2}\right)^{3/2}$$

$$\mathcal{Z} = \exp\left(z e^{\mu/RT}\right) = \exp\left[V\left(\frac{2\pi m k_{\mathrm{B}} T}{h^2}\right)^{3/2} e^{\mu/RT}\right]$$

Hence,

$$\phi = -k_{\mathrm{B}} T \ln \mathcal{Z} = -k_{\mathrm{B}} T V\left(\frac{2\pi m k_{\mathrm{B}} T}{h^2}\right)^{3/2} e^{\mu/RT}$$

Example: 3D Ideal Gas

$$\phi = -k_{\mathrm{B}}TV\left(\frac{2\pi m k_{\mathrm{B}}T}{h^2}\right)^{3/2} e^{\mu/RT}$$

$$n = -\left(\frac{\partial\phi}{\partial\mu}\right)_{T,V} = k_{\mathrm{B}}TV\left(\frac{2\pi m k_{\mathrm{B}}T}{h^2}\right)^{3/2}\frac{e^{\mu/RT}}{RT} = -\frac{\phi}{RT}$$

$$p = -\left(\frac{\partial\phi}{\partial V}\right)_{T,\mu} = k_{\mathrm{B}}T\left(\frac{2\pi m k_{\mathrm{B}}T}{h^2}\right)^{3/2} e^{\mu/RT} = -\frac{\phi}{V}$$

$$\therefore \phi = -pV = -nRT$$

Quantum Particles

We only know the number of particles in microstate i

Consider two particles in three single-particle microstates:

- Distinguishable particles: $W = 3 \times 3 = 9$
- Indistinguishable particles: $W = 9/2! = 4.5$
- Quantum particles: for (n_A, n_B, n_C),

  (1, 1, 0), (1, 0, 1), (0, 1, 1), (2, 0, 0), (0, 2, 0), (0, 0, 2) $\rightarrow W = 6$

The Gibbs correction factor $1/N!$ is not 100% accurate

Bosons and Fermions

- Bosons: particles that can share a one-particle microstate with another of the same species
  - *e.g.* photon, helium-4 nucleus, …
- Fermions: particles that cannot share a one-particle microstate with another of the same species
  - *e.g.* electron, proton, neutron, helium-3 nucleus …
  - Pauli exclusion principle: two identical fermions cannot occupy the same state

Occupation Numbers

For each single-particle state k, suppose that:

1) We have N_k identical particles.

2) Each particle has energy ϵ_k.

Then the system energy and particle numbers are

$$E = \sum_k N_k \epsilon_k \, ; \qquad N = \sum_k N_k$$

What is the average occupation number of a single-particle state with energy ϵ?

(Distinguishable) Classical Particles

Probability of finding a single particle in state with energy ϵ:

$$p = \frac{e^{-\epsilon/k_{\mathrm{B}}T}}{z}$$

The average occupation number is

$$N_{\mathrm{MB}}(\epsilon) = Np = \frac{N}{z} e^{-\epsilon/k_{\mathrm{B}}T}$$

However, we derived $\mu = -RT \ln(z/N) = N_{\mathrm{A}}\tilde{\mu}$;

$$\therefore N_{\mathrm{MB}}(\epsilon) = e^{\tilde{\mu}/k_{\mathrm{B}}T} e^{-\epsilon/k_{\mathrm{B}}T} = e^{-(\epsilon-\tilde{\mu})/k_{\mathrm{B}}T}$$

Grand Partition Function

$$\mathcal{Z} = \sum_i e^{-E_i/k_\mathrm{B}T + \tilde{\mu}N_i/k_\mathrm{B}T} = \sum_{\{N_k\}} e^{-\sum_k N_k(\epsilon_k - \tilde{\mu})/k_\mathrm{B}T}$$

$$= \sum_{\{N_k\}} \prod_k e^{-N_k(\epsilon_k - \tilde{\mu})/k_\mathrm{B}T} = \prod_k \sum_{N_k} e^{-N_k(\epsilon_k - \tilde{\mu})/k_\mathrm{B}T}$$

$$\mathcal{Z}_k = \sum_{N_k} e^{-N_k(\epsilon_k - \tilde{\mu})/k_\mathrm{B}T}$$

$$\mathcal{Z} = \prod_k \mathcal{Z}_k$$

Evaluating Summation

$$\mathcal{Z}_k = \sum_{N_k} e^{-N_k(\epsilon_k - \tilde{\mu})/k_\mathrm{B}T}$$

The average occupation number of state k is

$$\langle N_k(\epsilon_k)\rangle = \sum_{N_k} N_k p(N_k) = \sum_{N_k} N_k \frac{e^{-N_k(\epsilon_k - \tilde{\mu})/k_\mathrm{B}T}}{\mathcal{Z}}$$

- Fermion: $N_k = 0$ or 1
- Boson: $N_k =$ any nonnegative integer (0, 1, 2, …)

Fermions

$$\mathcal{Z}_k = \sum_{N_k} e^{-N_k(\epsilon_k - \tilde{\mu})/k_\mathrm{B}T}$$

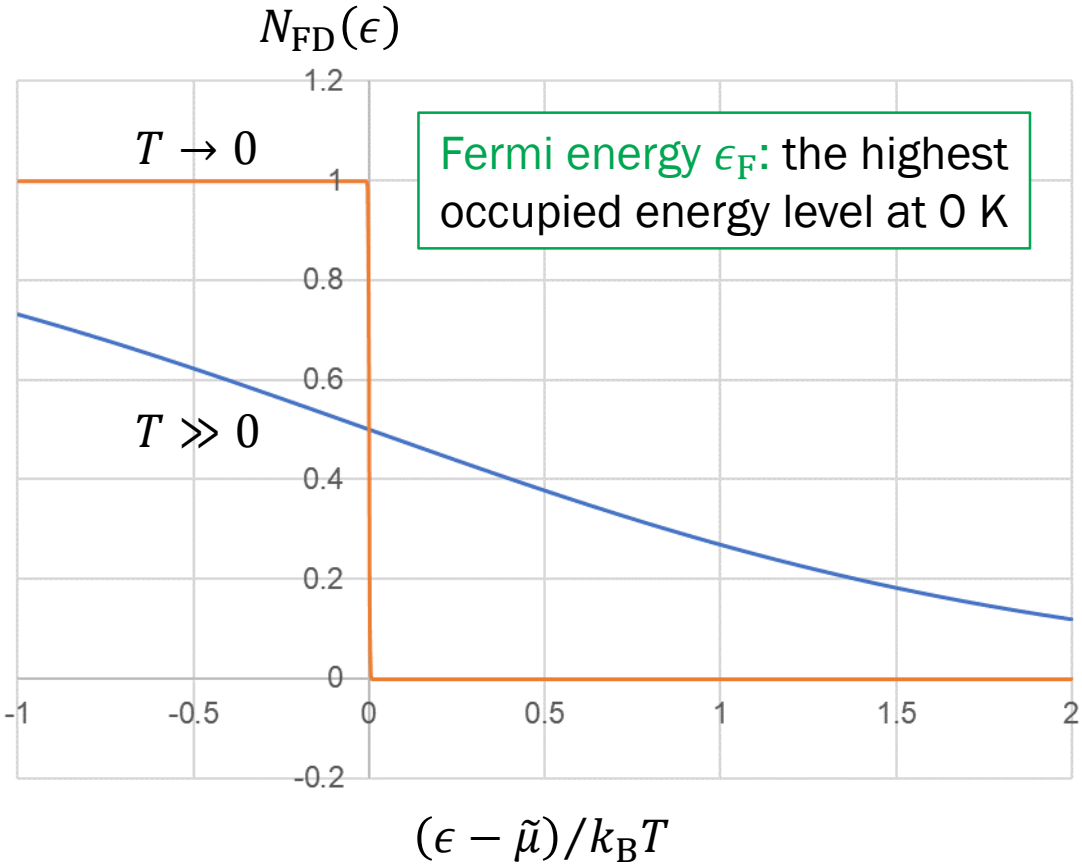
For a specific microstate, only $N_k = 0$ or $N_k = 1$ is possible:

$$\mathcal{Z}_k = 1 + e^{-(\epsilon_k - \tilde{\mu})/k_\mathrm{B}T}$$

The average occupation number is

$$\langle N_k(\epsilon_k) \rangle = \sum_{N_k=0,1} N_k p(N_k) = \frac{e^{-(\epsilon_k - \tilde{\mu})/k_\mathrm{B}T}}{1 + e^{-(\epsilon_k - \tilde{\mu})/k_\mathrm{B}T}} = \frac{1}{e^{(\epsilon_k - \tilde{\mu})/k_\mathrm{B}T} + 1}$$

Fermi Energy



Fermi energy ϵ_{F}: the highest occupied energy level at 0 K

Bosons

$$\mathcal{Z}_k = \sum_{N_k} e^{-N_k(\epsilon_k - \tilde{\mu})/k_\mathrm{B}T} = \sum_{N_k} e^{-N_k x}$$

by letting $x = (\epsilon_k - \tilde{\mu})/k_\mathrm{B}T$.

Now, $N_k = 0, 1, 2, \cdots$ is possible:

$$\mathcal{Z}_k = 1 + e^{-x} + e^{-2x} + \cdots$$

Using the closed form of the geometric series,

$$\mathcal{Z}_k = \frac{1}{1 - e^{-x}}$$

Mathematical Trick

$$\mathcal{Z}_k = \sum_{N_k} e^{-N_k(\epsilon_k - \tilde{\mu})/k_\mathrm{B}T} = \sum_{N_k} e^{-N_k x}$$

$$\langle N_k(\epsilon_k)\rangle = \sum_{N_k} \frac{N_k e^{-N_k(\epsilon_k - \tilde{\mu})/k_\mathrm{B}T}}{\mathcal{Z}_k} = \sum_{N_k} \frac{N_k e^{-N_k x}}{\mathcal{Z}_k}$$

$$\frac{\partial(\ln \mathcal{Z}_k)}{\partial x} = \frac{1}{\mathcal{Z}_k}\frac{\partial \mathcal{Z}_k}{\partial x} = -\frac{1}{\mathcal{Z}_k}\sum_{N_k} N_k e^{-N_k x} = -\langle N_k(\epsilon_k)\rangle$$

Bosons

$$\mathcal{Z}_k = \frac{1}{1 - e^{-x}}$$

$$\langle N_k(\epsilon_k) \rangle = -\frac{\partial(\ln \mathcal{Z}_k)}{\partial x} = \frac{\partial}{\partial x} \ln(1 - e^{-x}) = \frac{e^{-x}}{1 - e^{-x}}$$

Hence,

$$\langle N_k(\epsilon_k) \rangle = \frac{1}{e^x - 1} = \frac{1}{e^{(\epsilon_k - \tilde{\mu})/k_B T} - 1}$$

Average Occupation Number

For a specific microstate with energy ϵ,

- Classical particles (Maxwell-Boltzmann distribution)

$$N_{\mathrm{MB}}(\epsilon) = e^{-(\epsilon-\tilde{\mu})/k_{\mathrm{B}}T}$$

- Fermions (Fermi-Dirac distribution)

$$N_{\mathrm{FD}}(\epsilon) = \frac{1}{e^{(\epsilon-\tilde{\mu})/k_{\mathrm{B}}T} + 1}$$

- Bosons (Bose-Einstein distribution)

$$N_{\mathrm{BE}}(\epsilon) = \frac{1}{e^{(\epsilon-\tilde{\mu})/k_{\mathrm{B}}T} - 1}$$

Average Occupation Number

